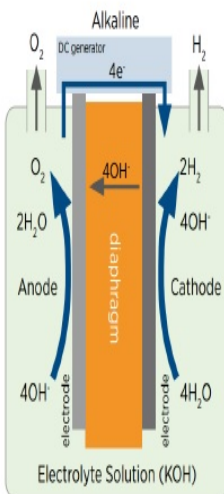
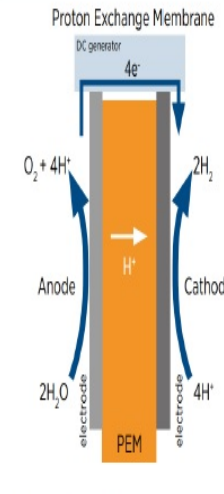
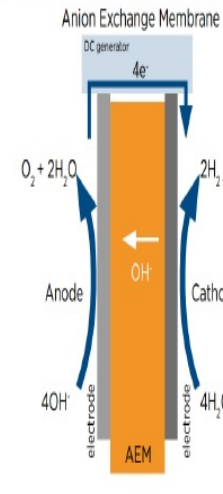
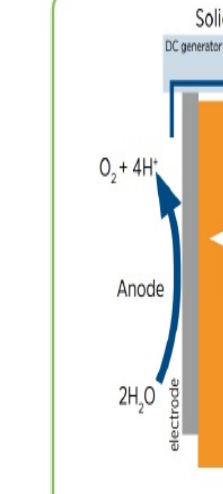


Qualification Pack

Low Temperature Electrolyser (< 100 °C)		High Temperature (> 800 °C)
Alkaline Electrolysis	Acidic Electrolysis	Solid Oxide Ceramic membrane
Membrane Free	Solid Polymer Electrolyte Membrane	

Alkaline  Anode: $4\text{OH}^- \leftrightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ Cathode: $4\text{H}_2\text{O} + 4\text{e}^- \leftrightarrow 2\text{H}_2 + 4\text{OH}^-$	Proton Exchange Membrane  Anode: $2\text{H}_2\text{O} \leftrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ Cathode: $4\text{H}^+ + 4\text{e}^- \leftrightarrow 2\text{H}_2$	Anion Exchange Membrane  Anode: $4\text{OH}^- \leftrightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ Cathode: $4\text{H}_2\text{O} + 4\text{e}^- \leftrightarrow 2\text{H}_2 + 4\text{OH}^-$	Solid Oxide  Anode: $2\text{O}^{2-} \leftrightarrow \text{O}_2 + 4\text{e}^-$ Cathode: $2\text{H}_2\text{O} + 4\text{e}^- \leftrightarrow 2\text{H}_2 + 2\text{O}^{2-}$
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Green Hydrogen Plant Junior Technician-Electrolyzer

QP Code: SGJ/Q4302

Version: 1.0

NSQF Level: 3

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi - 110021 || email: Rupali Sinha

Qualification Pack

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Qualification Pack

SGJ/Q4302: Green Hydrogen Plant Junior Technician-Electrolyzer

Brief Job Description

Green Hydrogen Plant Junior Technician-Electrolyzer would primarily install electrolyzer units along with other Mechanical and Electrical equipment of a green hydrogen plant and perform the associated civil works. He/She would also assist in performing Pre-commissioning checks/tests and install & commission electrolyzer unit along with other sub system in a green hydrogen/green ammonia plant.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete the work safely. He/she should be able to communicate in local language. He / She must possess energy and strength for physical work. And also have a basic understanding of social and natural environment. He/she should possess very good interpersonal skills to work in electrolyser of green hydrogen plant.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4301: Basics of Green Hydrogen Production](#)
2. [SGJ/N4302: Analyse Main Parts of green hydrogen production unit](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
4. [SGJ/N4306: Installation of Electrolyzer for green hydrogen production plant](#)
5. [SGJ/N4307: Perform Operation and Maintenance of Electrolyzer](#)
6. [SGJ/N4048: Perform Health and safety measures for installing and operating Electrolyser at Green hydrogen Plant](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
Country	India
NSQF Level	3

Qualification Pack

Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8131.2100
Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR 8th grade pass (with two year of (NTC/ NAC) after 8th) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 2.5) with 1-2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-03-ES-00765-2023-V1-SCGJ
NQR Version	1

Remarks:

Total 330 Hours i.e. 11 Credits (Theory: 130 hours+Practical:110 hours+ 30 hours of employability skills + 60 hours of OJT)

Qualification Pack

SGJ/N4301: Basics of Green Hydrogen Production

Description

This unit specified the fundamentals of green hydrogen production system

Scope

The scope covers the following :

- Basics of Green Hydrogen Production

Elements and Performance Criteria

Basics of Green Hydrogen Production

To be competent, the user/individual on the job must be able to:

- PC1.** discuss properties and characteristics of Hydrogen
- PC2.** describe basic concepts of Hydrogen as fuel and energy carrier
- PC3.** discuss in brief various existing methods of hydrogen production and demonstrate various methods of hydrogen production.
- PC4.** discuss various colour code nomenclature of Hydrogen and demonstrate with chart colour code nomenclature of Hydrogen
- PC5.** discuss various technology options for production of Green Hydrogen and draw a flow diagram of green hydrogen production, conversion and end uses across the energy system
- PC6.** discuss key aspects and challenges related to production and storage of Green Hydrogen and do an activity for matching the process and source of production as applicable for different colour codes of hydrogen
- PC7.** discuss the end use applications of Green hydrogen in industry, transport and power production
- PC8.** discuss the role and responsibilities of various Green Hydrogen Plant Junior Technicians and demonstrate roles of various technicians in a green hydrogen plant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizations reporting structure
- KU2.** organizations documentation policy
- KU3.** organizational culture
- KU4.** Properties and characteristic of hydrogen
- KU5.** concept of hydrogen as fuel and energy carrier
- KU6.** signs, symbols and color codes and nomenclature used in hydrogen production
- KU7.** challenges related to production and storage of green hydrogen
- KU8.** basic skills required to perform the task of green hydrogen production

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up relevant documents.
- GS2.** read vernacular language.
- GS3.** understand the various colour codes used in hydrogen production
- GS4.** express statements or information clearly so that others can understand
- GS5.** understand the main points of simple discussions
- GS6.** follow organization rule-based decision making process.
- GS7.** planning and organization of work to meet schedule.
- GS8.** work constructively and collaboratively with others.
- GS9.** communicate and create awareness.
- GS10.** recognize problems & approach relevant authority.
- GS11.** critically evaluate information obtained from supervisor and co-workers to perform day to day activities.
- GS12.** ask questions for better understanding.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Green Hydrogen Production</i>	30	20	-	-
PC1. discuss properties and characteristics of Hydrogen	3	-	-	-
PC2. describe basic concepts of Hydrogen as fuel and energy carrier	3	-	-	-
PC3. discuss in brief various existing methods of hydrogen production and demonstrate various methods of hydrogen production.	4	2	-	-
PC4. discuss various colour code nomenclature of Hydrogen and demonstrate with chart colour code nomenclature of Hydrogen	4	3	-	-
PC5. discuss various technology options for production of Green Hydrogen and draw a flow diagram of green hydrogen production, conversion and end uses across the energy system	4	5	-	-
PC6. discuss key aspects and challenges related to production and storage of Green Hydrogen and do an activity for matching the process and source of production as applicable for different colour codes of hydrogen	4	5	-	-
PC7. discuss the end use applications of Green hydrogen in industry, transport and power production	4	-	-	-
PC8. discuss the role and responsibilities of various Green Hydrogen Plant Junior Technicians and demonstrate roles of various technicians in a green hydrogen plant	4	5	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4301
NOS Name	Basics of Green Hydrogen Production
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4302: Analyse Main Parts of green hydrogen production unit

Description

This unit specifies identifying the main parts of green hydrogen production unit

Scope

The scope covers the following :

- Analyse Main Parts of green hydrogen production unit

Elements and Performance Criteria

Analyse Main Parts of green hydrogen production unit

To be competent, the user/individual on the job must be able to:

- PC1.** identify key parts and components of the Green Hydrogen plant including electrical, mechanical and civil components and illustrate the schematic of Green hydrogen production plant
- PC2.** discuss functions of each part and components and illustrate key components of the plant and outline their functions through plant schematic
- PC3.** discuss fundamental principles of main components on which they operate
- PC4.** explain basics of plant layout and illustrate how to interpret the Plant Layout including various equipments
- PC5.** discuss the plant components most relevant to the course and demonstrate how to interpret signs, notices and/or cautions at project site.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizations reporting structure
- KU2.** organizations documentation policy
- KU3.** organizational culture
- KU4.** Properties and characteristic of hydrogen
- KU5.** concept of hydrogen as fuel and energy carrier
- KU6.** signs, symbols and color codes and nomenclature used in hydrogen production
- KU7.** challenges related to production and storage of green hydrogen
- KU8.** basic skills required to perform the task of green hydrogen production
- KU9.** understanding relevant regulations and safety standards

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** fill up relevant documents
- GS2.** read vernacular language
- GS3.** understand the various colour codes used in hydrogen production
- GS4.** express statements or information clearly so that others can understand
- GS5.** understand the main points of simple discussions
- GS6.** follow organization rule-based decision making process
- GS7.** planning and organization of work to meet schedule
- GS8.** work constructively and collaboratively with others
- GS9.** communicate and create awareness
- GS10.** ask questions for better understanding.
- GS11.** critically evaluate information obtained from supervisor and co-workers to perform day to day activities.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse Main Parts of green hydrogen production unit</i>	30	20	-	-
PC1. identify key parts and components of the Green Hydrogen plant including electrical, mechanical and civil components and illustrate the schematic of Green hydrogen production plant	6	5	-	-
PC2. discuss functions of each part and components and illustrate key components of the plant and outline their functions through plant schematic	6	5	-	-
PC3. discuss fundamental principles of main components on which they operate	6	5	-	-
PC4. explain basics of plant layout and illustrate how to interpret the Plant Layout including various equipments	6	-	-	-
PC5. discuss the plant components most relevant to the course and demonstrate how to interpret signs, notices and/or cautions at project site.	6	5	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4302
NOS Name	Analyse Main Parts of green hydrogen production unit
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

Qualification Pack

SGJ/N4306: Installation of Electrolyzer for green hydrogen production plant

Description

This unit explain about Installation of Electrolyzer for green hydrogen production plant

Scope

The scope covers the following :

- About Electrolyser Technology
- types of Electrolyzer, their broad specifications and comparison
- step by step process of installation of Electrolyser
- equipment required for installation, commissioning, dismantling and removal of components of Electrolyser

Elements and Performance Criteria

Fundamental of Electrolyzer technology

To be competent, the user/individual on the job must be able to:

- PC1.** provide an overview of various electrolyser along with their key features and comparison and illustrate different types of Electrolyser and their key specification through Pictures, videos, product data sheet etc
- PC2.** explain the major components of an Electrolyser
- PC3.** discuss the Operating principles of PEM, AE and SOEC electrolyser and outline differences in PEM, AE, Electrolyser and illustrate their schematics
- PC4.** explain the inputs / outputs of an Electrolyser system
- PC5.** discuss basic technical specifications of electrolyser
- PC6.** discuss the concept of electrolyser technology for Green Hydrogen generation in India and overseas and showcase the successful case studies of electrolyser technology for Green Hydrogen generation in India and overseas

Perform Installation of Electrolyser for green hydrogen generation

To be competent, the user/individual on the job must be able to:

- PC7.** identify tools and equipment required for Installation of electrolyser, Plant & Machineries conforming to relevant technical sheets, safety and technical standards for proper execution of work and show how how to identify the tools and equipment to perform the installation of Electrolyser as per concerned technical sheets
- PC8.** discuss the assembly/Installation of Parts and Components of Electrolyzer and show how to handle different tools, equipment as per concerned standard and industry practices.
- PC9.** discuss how to perform stack assembling, and its testing and show how how to monitor various parameters of electrolysis
- PC10.** discuss the importance of Manifold and sealing

Qualification Pack

PC11. identify the key learnings of successful installation of Electrolyser projects for green hydrogen projects in India and overseas and showcase the key aspects of successful installation of Electrolyser projects for green hydrogen projects in India and overseas.

PC12. show how to perform requisite data and document management

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizations reporting structure
- KU2.** organizations documentation policy
- KU3.** organizational culture
- KU4.** Properties and characteristic of hydrogen
- KU5.** concept of hydrogen as fuel and energy carrier
- KU6.** signs, symbols and color codes and nomenclature used in hydrogen production
- KU7.** understanding relevant regulations and safety standards
- KU8.** challenges related to production and storage of green hydrogen
- KU9.** basic skills required to perform the task of green hydrogen production
- KU10.** specific requirements relating to hydrogen infrastructure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read vernacular/English language
- GS2.**
 - participate in and understand the main points of discussions with teams and external
 - stakeholders
- GS3.** express statements or information clearly so that others can understand
- GS4.** plan and organize work schedule to meet work deadlines
- GS5.** take decision with systematic course of actions and/or response
- GS6.** read and write different types of documents/instructions/correspondence
- GS7.** communicate effectively using appropriate language in formal and informal settings
- GS8.** work constructively and collaboratively with others.
- GS9.** communicate and create awareness
- GS10.**
 - critically evaluate information obtained from supervisor and co-workers to perform day to
 - day activities.
- GS11.** planning and organization of work to meet schedule.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental of Electrolyzer technology</i>	17	10	-	-
PC1. provide an overview of various electrolyser along with their key features and comparison and illustrate different types of Electrolyser and their key specification through Pictures, videos, product data sheet etc	2	3	-	-
PC2. explain the major components of an Electrolyser	3	-	-	-
PC3. discuss the Operating principles of PEM, AE and SOEC electrolyser and outline differences in PEM, AE, Electrolyser and illustrate their schematics	2	3	-	-
PC4. explain the inputs / outputs of an Electrolyser system	4	-	-	-
PC5. discuss basic technical specifications of electrolyser	4	-	-	-
PC6. discuss the concept of electrolyser technology for Green Hydrogen generation in India and overseas and showcase the successful case studies of electrolyser technology for Green Hydrogen generation in India and overseas	2	4	-	-
<i>Perform Installation of Electrolyser for green hydrogen generation</i>	11	12	-	-
PC7. identify tools and equipment required for Installation of electrolyser, Plant & Machineries conforming to relevant technical sheets, safety and technical standards for proper execution of work and show how to identify the tools and equipment to perform the installation of Electrolyser as per concerned technical sheets	2	2	-	-
PC8. discuss the assembly/Installation of Parts and Components of Electrolyzer and show how to handle different tools, equipment as per concerned standard and industry practices.	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. discuss how to perform stack assembling, and its testing and show how how to monitor various parameters of electrolysis	2	2	-	-
PC10. discuss the importance of Manifold and sealing	2	-	-	-
PC11. identify the key learnings of successful installation of Electrolyser projects for green hydrogen projects in India and overseas and showcase the key aspects of successful installation of Electrolyser projects for green hydrogen projects in India and overseas.	3	3	-	-
PC12. show how to perform requisite data and document management	-	3	-	-
NOS Total	28	22	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4306
NOS Name	Installation of Electrolyzer for green hydrogen production plant
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4307: Perform Operation and Maintenance of Electrolyzer

Description

This unit explain and Perform Operation and Maintenance of Electrolyzer

Scope

The scope covers the following :

- process of electrolyser unit operation
- maintenance of Electrolyser
- replaces and removal of components

Elements and Performance Criteria

Perform Operation and Maintenance of Electrolyzer

To be competent, the user/individual on the job must be able to:

- PC1.** explain the layout of electrolyser unit and illustrate the schematic of Electrolyser Unit
- PC2.** discuss the different operating conditions of Electrolyser like, Input power condition, temperature and pressure conditions and illustrate the schematic of input power to electrolyser and the operation of the system
- PC3.** explain Start up, shutdown, and operate the production processes and prepare the standard operating procedure(sop) for Startup, shutdown, and operation procedure of the electrolyser
- PC4.** discuss the reasons for Never operate the unit in below-zero temperatures and its causes
- PC5.** discuss about the common problems of electrolyser unit during operation of electrolyser unit and show how to follow Standard operating procedure of the electrolyser
- PC6.** discuss the preventive maintenance plan and explain the activities carried out during routine maintenance work and prepare the standard maintenance practice (SMP) for the maintenance of electrolyser
- PC7.** discuss checking and detection of leakage of hydrogen
- PC8.** discuss how to purge the hydrogen out of the equipment during maintenance and show how to safely purge the hydrogen out of the equipment during maintenance
- PC9.**
 - discuss general Step for repairing or replacing the equipment Includes-
 - preparing the system including isolating energy sources via Lockout/Tagout (LOTO) and purging hydrogen out of the equipment
 - Inspection
 - Doing the work
 - Leak testing
 - Purging air out of the equipment
- PC10.** discuss importance of checking the hydrogen purity.
- PC11.** discuss the routine inspection carried out to check the Stacks are corrosion free and showcase the do's and don'ts activity during operation and maintenance of electrolyser.
- PC12.** discuss the importance of Maintaining the electrolyser temperature as per the specification
- PC13.** discuss about the water quality feed to electrolyser for green hydrogen generation

Qualification Pack

- PC14.** discuss about routine maintenance of electrolyser and showcase the do's and don'ts activity during operation and maintenance of electrolyser.
- PC15.** discuss the fouling issue and its prevention depict the fouling issue in electrolyser through pictures and videos
- PC16.** discuss about working of all measuring instruments, controllers, and control valves.
- PC17.** discuss the function and attribute of pressure control function and other safety device functions.
- PC18.** discuss the routine check-up of input power supply to electrolyser and ensure no leakage current found.
- PC19.** identify the best Operation and maintenance practices for safe and efficient production of green hydrogen in India and overseas and analyse the best Operation and maintenance procedures for safe and efficient production of green hydrogen in India and overseas
- PC20.** show how to perform the procedure for electrolyser stack assembly and its maintenance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Layout of electrolyser unit
- KU2.** different operating condition of electrolyser like, Input power condition, temperature and pressure conditions
- KU3.** start up ,shutdown, operation and maintenance of electrolyser
- KU4.** preventive maintenance of electrolyser
- KU5.** Understanding relevant regulations and safety standards
- KU6.** specific requirements relating to hydrogen infrastructure

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read vernacular/English language
- GS2.** • participate in and understand the main points of discussions with teams and external
• stakeholders
- GS3.** express statements or information clearly so that others can understand
- GS4.** plan and organize work schedule to meet work deadlines
- GS5.** take decision with systematic course of actions and/or response
- GS6.** read and write different types of documents/instructions/correspondence
- GS7.** communicate effectively using appropriate language in formal and informal settings
- GS8.** behave politely and appropriately with all stakeholder
- GS9.** work constructively and collaboratively with others.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Operation and Maintenance of Electrolyzer</i>	32	18	-	-
PC1. explain the layout of electrolyser unit and illustrate the schematic of Electrolyser Unit	2	2	-	-
PC2. discuss the different operating conditions of Electrolyser like, Input power condition, temperature and pressure conditions and illustrate the schematic of input power to electrolyser and the operation of the system	1	2	-	-
PC3. explain Start up, shutdown, and operate the production processes and prepare the standard operating procedure(sop) for Startup, shutdown, and operation procedure of the electrolyser	1	2	-	-
PC4. discuss the reasons for Never operate the unit in below-zero temperatures and its causes	1	-	-	-
PC5. discuss about the common problems of electrolyser unit during operation of electrolyser unit and show how to follow Standard operating procedure of the electrolyser	1	2	-	-
PC6. discuss the preventive maintenance plan and explain the activities carried out during routine maintenance work and prepare the standard maintenance practice (SMP) for the maintenance of electrolyser	1	2	-	-
PC7. discuss checking and detection of leakage of hydrogen	1	-	-	-
PC8. discuss how to purge the hydrogen out of the equipment during maintenance and show how to safely purge the hydrogen out of the equipment during maintenance	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. - discuss general Step for repairing or replacing the equipment Includes- - preparing the system including isolating energy sources via Lockout/Tagout (LOTO) and purging hydrogen out of the equipment - Inspection - Doing the work - Leak testing - Purging air out of the equipment	2	-	-	-
PC10. discuss importance of checking the hydrogen purity.	2	-	-	-
PC11. discuss the routine inspection carried out to check the Stacks are corrosion free and showcase the do's and don'ts activity during operation and maintenance of electrolyser.	1	2	-	-
PC12. discuss the importance of Maintaining the electrolyser temperature as per the specification	2	-	-	-
PC13. discuss about the water quality feed to electrolyser for green hydrogen generation	2	-	-	-
PC14. discuss about routine maintenance of electrolyser and showcase the do's and don'ts activity during operation and maintenance of electrolyser.	2	-	-	-
PC15. discuss the fouling issue and its prevention depict the fouling issue in electrolyser through pictures and videos	2	2	-	-
PC16. discuss about working of all measuring instruments, controllers, and control valves.	2	-	-	-
PC17. discuss the function and attribute of pressure control function and other safety device functions.	2	-	-	-
PC18. discuss the routine check-up of input power supply to electrolyser and ensure no leakage current found.	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. identify the best Operation and maintenance practices for safe and efficient production of green hydrogen in India and overseas and analyse the best Operation and maintenance procedures for safe and efficient production of green hydrogen in India and overseas	2	2	-	-
PC20. show how to perform the procedure for electrolyser stack assembly and its maintenance	2	-	-	-
NOS Total	32	18	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4307
NOS Name	Perform Operation and Maintenance of Electrolyzer
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4048: Perform Health and safety measures for installing and operating Electrolyser at Green hydrogen Plant

Description

This unit explain about how to perform Health & Safety measures in Electrolyser unit at Green hydrogen Plant.

Scope

The scope covers the following :

- requirements for safe work area at Electrolyser in hydrogen generation project site.
- understand hazards associated with Electrolyser in a hydrogen generation system and their mitigation measures

Elements and Performance Criteria

Maintain Health & Safety at Electrolyser of Green Hydrogen generation project site

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area at Electrolyser in hydrogen generation project site.
- PC2.** explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of Electrolyser systems and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site
- PC3.** describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.
- PC4.** discuss importance of different detectors and safety tools
- PC5.** review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
- PC6.** explain the importance of administering first aid and demonstrate how to administer first aid.
- PC7.** identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work of Electrolyser
- PC8.** identify the hazards associated with Electrolyser and hydrogen generation system
- PC9.** identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations
- PC10.** discuss Mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system.
- PC11.** discuss all applicable statutory requirements along with safety regulations in terms of fire protection and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection
- PC12.** discuss how to incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.

Qualification Pack

- PC13.** identify best practices for implementing health and safety measures in installation and operations of electrolyser for green hydrogen production in India and overseas and showcase best practices for implementing health and safety measures in installation and operations of electrolyser for green hydrogen production in India and overseas.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** document information using appropriate corporate forms
- KU3.** obtain authorization from specified field safety officer and supervisor
- KU4.** relevant personal protective equipment required within the facility
- KU5.** relevant standards and regulations to be followed in the facility
- KU6.** occupational health and safety (OHS) standards to be followed in the facility
- KU7.** risk identification and mitigation procedure for safe work in in a facility
- KU8.** know how of tools & tackles required to carry out the work
- KU9.** understand the various colour codes used in hydrogen production
- KU10.** challenges related to production and storage of green hydrogen
- KU11.** properties and characteristic of hydrogen

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** organizations reporting structure
- GS2.** organizations documentation policy
- GS3.** organizational culture
- GS4.** signs, symbols and color codes and nomenclature used in hydrogen production
- GS5.** understanding relevant regulations and safety standards
- GS6.** basic skills required to perform the task of green hydrogen production
- GS7.** specific requirements relating to hydrogen infrastructure
- GS8.** express statements or information clearly so that others can understand
- GS9.** follow organization rule-based decision making process
- GS10.** planning and organization of work to meet schedule

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain Health & Safety at Electrolyser of Green Hydrogen generation project site</i>	30	20	-	-
PC1. explain the requirements for safe work area at Electrolyser in hydrogen generation project site.	2	2	-	-
PC2. explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of Electrolyser systems and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen generation site	2	2	-	-
PC3. describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.	2	-	-	-
PC4. discuss importance of different detectors and safety tools	2	-	-	-
PC5. review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	2	2	-	-
PC6. explain the importance of administering first aid and demonstrate how to administer first aid.	3	2	-	-
PC7. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work of Electrolyser	3	2	-	-
PC8. identify the hazards associated with Electrolyser and hydrogen generation system	2	-	-	-
PC9. identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations	2	-	-	-
PC10. discuss Mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system.	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. discuss all applicable statutory requirements along with safety regulations in terms of fire protection and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection	2	2	-	-
PC12. discuss how to incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.	3	3	-	-
PC13. identify best practices for implementing health and safety measures in installation and operations of electrolyser for green hydrogen production in India and overseas and showcase best practices for implementing health and safety measures in installation and operations of electrolyser for green hydrogen production in India and overseas.	3	3	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4048
NOS Name	Perform Health and safety measures for installing and operating Electrolyser at Green hydrogen Plant
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4301.Basics of Green Hydrogen Production	30	20	0	0	50	15
SGJ/N4302.Analyse Main Parts of green hydrogen production unit	30	20	0	0	50	17
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	17
SGJ/N4306.Installation of Electrolyzer for green hydrogen production plant	28	22	0	0	50	17
SGJ/N4307.Perform Operation and Maintenance of Electrolyzer	32	18	0	0	50	17
SGJ/N4048.Perform Health and safety measures for installing and operating Electrolyser at Green hydrogen Plant	30	20	0	0	50	17
Total	170	130	-	-	300	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.